

Gaussian Elimination Method

Gaussian Elimination Method

System of Linear Equations:

$$\mathbf{Ax} = \mathbf{b}$$
$$\begin{bmatrix} a_{11} & a_{12} & a_{13} & \cdots & a_{1n} \\ a_{21} & a_{22} & a_{23} & \cdots & a_{2n} \\ a_{31} & a_{32} & a_{33} & \cdots & a_{3n} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & a_{n3} & \cdots & a_{nn} \end{bmatrix} \begin{Bmatrix} x_1 \\ x_2 \\ x_3 \\ \vdots \\ x_n \end{Bmatrix} = \begin{Bmatrix} b_1 \\ b_2 \\ b_3 \\ \vdots \\ b_n \end{Bmatrix}$$

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Forward Elimination:

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$$\begin{bmatrix} a_{11} & a_{12} & a_{13} & \cdots & a_{1n} \\ 0 & a'_{22} & a'_{23} & \cdots & a'_{2n} \\ 0 & 0 & a''_{33} & \cdots & a''_{3n} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & a''_{n3} & \cdots & a''_{nn} \end{bmatrix} \begin{Bmatrix} x_1 \\ x_2 \\ x_3 \\ \vdots \\ x_n \end{Bmatrix} = \begin{Bmatrix} b_1 \\ b'_2 \\ b''_3 \\ \vdots \\ b''_n \end{Bmatrix}$$

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Back Substitution:

$$x_n = \frac{b^{(n-1)}_n}{a^{(n-1)}_{nn}}$$

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Back Substitution:

$$x_n = \frac{b^{(n-1)}_n}{a^{(n-1)}_{nn}} \rightarrow x_i = \frac{b^{(i-1)}_i - \sum_{j=i+1}^n a^{(i-1)}_{ij} \cdot x_j}{a^{(i-1)}_{ii}}; \quad i = n-1, n-2, \dots, 1$$

Gaussian Elimination Method

Example 03-01: Find root of the following system of equations.

$$E_1: \quad x_1 - x_2 + 2x_3 - x_4 = -8$$

$$E_2: \quad 2x_1 - 2x_2 + 3x_3 - 3x_4 = -20$$

$$E_3: \quad x_1 + x_2 + x_3 = -2$$

$$E_4: \quad x_1 - x_2 + 4x_3 + 3x_4 = 4$$

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Forward Elimination:

Gaussian Elimination Method

Example 03-01: Find root of the following system of equations.

$$\begin{array}{rclclclcl} E_1: & x_1 & - & x_2 & + & 2x_3 & - & x_4 & = & -8 \\ E_2: & 2x_1 & - & 2x_2 & + & 3x_3 & - & 3x_4 & = & -20 \\ E_3: & x_1 & + & x_2 & + & x_3 & & & = & -2 \\ E_4: & x_1 & - & x_2 & + & 4x_3 & + & 3x_4 & = & 4 \end{array}$$

Forward Elimination:

$$E'_2 = E_2 - (a_{21}/a_{11}) \times E_1:$$

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$$(\quad 2x_1 - 2x_2 + 3x_3 - 3x_4 = -20 \quad)$$

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Forward Elimination:

$$E'_2 = E_2 - (a_{21}/a_{11}) \times E_1:$$
$$\begin{array}{l} \quad \quad \quad (\quad 2x_1 - 2x_2 + 3x_3 - 3x_4 = -20 \quad) \\ -\frac{2}{1} \times (\quad x_1 - x_2 + 2x_3 - x_4 = -8 \quad) \end{array}$$

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Example 03-01: Find root of the following system of equations.

$$\begin{array}{rclclclclcl} E_1: & x_1 & - & x_2 & + & 2x_3 & - & x_4 & = & -8 \\ E_2: & 2x_1 & - & 2x_2 & + & 3x_3 & - & 3x_4 & = & -20 \\ E_3: & x_1 & + & x_2 & + & x_3 & & & = & -2 \\ E_4: & x_1 & - & x_2 & + & 4x_3 & + & 3x_4 & = & 4 \end{array}$$

Forward Elimination:

$$E'_2 = E_2 - (a_{21}/a_{11}) \times E_1: \quad \begin{array}{rclclclclcl} & (& 2x_1 & - & 2x_2 & + & 3x_3 & - & 3x_4 & = & -20 &) \\ -\frac{2}{1} \times & (& x_1 & - & x_2 & + & 2x_3 & - & x_4 & = & -8 &) \\ \hline & & & & - & x_3 & - & x_4 & = & -4 & \end{array}$$

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Forward Elimination:

$$E'_2 = E_2 - (a_{21}/a_{11}) \times E_1: \quad \begin{array}{rclclclclcl} & (& 2x_1 & - & 2x_2 & + & 3x_3 & - & 3x_4 & = & -20 &) \\ -\frac{2}{1} \times & (& x_1 & - & x_2 & + & 2x_3 & - & x_4 & = & -8 &) \\ \hline & & & & - & x_3 & - & x_4 & = & -4 & \end{array}$$

$$E'_3 = E_3 - (a_{31}/a_{11}) \times E_1:$$

Gaussian Elimination Method

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$$\begin{array}{rclclclclcl} E_1: & x_1 & - & x_2 & + & 2x_3 & - & x_4 & = & -8 \\ E_2: & 2x_1 & - & 2x_2 & + & 3x_3 & - & 3x_4 & = & -20 \\ E_3: & x_1 & + & x_2 & + & x_3 & & & = & -2 \\ E_4: & x_1 & - & x_2 & + & 4x_3 & + & 3x_4 & = & 4 \end{array}$$

Forward Elimination:

$$E'_2 = E_2 - (a_{21}/a_{11}) \times E_1: \quad \begin{array}{rclclclclcl} & (& 2x_1 & - & 2x_2 & + & 3x_3 & - & 3x_4 & = & -20 &) \\ -\frac{2}{1} \times & (& x_1 & - & x_2 & + & 2x_3 & - & x_4 & = & -8 &) \\ \hline & & & & & & - & x_3 & - & x_4 & = & -4 \\ \hline \hline & (& x_1 & + & x_2 & + & x_3 & & & = & -2 &) \end{array}$$

$$E'_3 = E_3 - (a_{31}/a_{11}) \times E_1:$$

Example 03-01: Find root of the following system of equations.

$$\begin{array}{rcccccccl} E_1: & x_1 & - & x_2 & + & 2x_3 & - & x_4 & = & -8 \\ E_2: & 2x_1 & - & 2x_2 & + & 3x_3 & - & 3x_4 & = & -20 \\ E_3: & x_1 & + & x_2 & + & x_3 & & & = & -2 \\ E_4: & x_1 & - & x_2 & + & 4x_3 & + & 3x_4 & = & 4 \end{array}$$

$$E'_2 = E_2 - (a_{21}/a_{11}) \times E_1: \quad \begin{array}{r} \begin{array}{cccccc} (& 2x_1 & - & 2x_2 & + & 3x_3 & - & 3x_4 & = & -20 &) \\ \textcolor{red}{-\frac{2}{1}} \times & (& x_1 & - & x_2 & + & 2x_3 & - & x_4 & = & -8 &) \\ \hline & & & & - & x_3 & - & x_4 & = & -4 & \\ \hline \end{array} \\ \\ \begin{array}{cccccc} (& x_1 & + & x_2 & + & x_3 & & & = & -2 &) \\ \textcolor{red}{-\frac{1}{1}} \times & (& x_1 & - & x_2 & + & 2x_3 & - & x_4 & = & -8 &) \end{array} \end{array}$$

Example 03-01: Find root of the following system of equations.

$$\begin{array}{rcccccccl} E_1: & x_1 & - & x_2 & + & 2x_3 & - & x_4 & = & -8 \\ E_2: & 2x_1 & - & 2x_2 & + & 3x_3 & - & 3x_4 & = & -20 \\ E_3: & x_1 & + & x_2 & + & x_3 & & & = & -2 \\ E_4: & x_1 & - & x_2 & + & 4x_3 & + & 3x_4 & = & 4 \end{array}$$

$$E'_2 = E_2 - (a_{21}/a_{11}) \times E_1:$$
$$\begin{array}{r} \left(\begin{array}{ccccccccc} & 2x_1 & - & 2x_2 & + & 3x_3 & - & 3x_4 & = & -20 \\ & x_1 & - & x_2 & + & 2x_3 & - & x_4 & = & -8 \end{array} \right) \\ \hline \qquad\qquad\qquad - \quad x_3 \quad - \quad x_4 \quad = \quad -4 \\ \hline \hline \end{array}$$
$$E'_3 = E_3 - (a_{31}/a_{11}) \times E_1:$$
$$\begin{array}{r} \left(\begin{array}{ccccccccc} & x_1 & + & x_2 & + & x_3 & & & = & -2 \\ & x_1 & - & x_2 & + & 2x_3 & - & x_4 & = & -8 \end{array} \right) \\ \hline \qquad\qquad\qquad 2x_2 \quad - \quad x_3 \quad + \quad x_4 \quad = \quad 6 \\ \hline \hline \end{array}$$

Gaussian Elimination Method

Example 03-01: Find root of the following system of equations.

$E_1:$

x_1

$-$

x_2

$+$

$2x_3$

$-$

x_4

$=$

-8

$E_2:$

$2x_1$

$-$

$2x_2$

$+$

$3x_3$

$-$

$3x_4$

$=$

-20

$E_3:$

x_1

$+$

x_2

$+$

x_3

$=$

-2

$E_4:$

x_1

$-$

x_2

$+$

$4x_3$

$+$

$3x_4$

$=$

4

Forward Elimination:

$E'_2 = E_2 - (a_{21}/a_{11}) \times E_1:$

$-\frac{2}{1} \times$

$($

$2x_1$

$-$

$2x_2$

$+$

$3x_3$

$-$

$3x_4$

$=$

-20

$)$

$($

x_1

$-$

x_2

$+$

$2x_3$

$-$

x_4

$=$

-8

$)$

$-$

x_3

$-$

x_4

$=$

-4

$E'_3 = E_3 - (a_{31}/a_{11}) \times E_1:$

$-\frac{1}{1} \times$

$($

x_1

$+$

x_2

$+$

x_3

$=$

-2

$)$

$($

x_1

$-$

x_2

$+$

$2x_3$

$-$

x_4

$=$

-8

$)$

$2x_2$

$-$

x_3

$+$

x_4

$=$

6

$E'_4 = E_4 - (a_{41}/a_{11}) \times E_1:$

Gaussian Elimination Method

Example 03-01: Find root of the following system of equations.

$E_1:$

x_1

$-$

x_2

$+$

$2x_3$

$-$

x_4

$=$

-8

$E_2:$

$2x_1$

$-$

$2x_2$

$+$

$3x_3$

$-$

$3x_4$

$=$

-20

$E_3:$

x_1

$+$

x_2

$+$

x_3

$=$

-2

$E_4:$

x_1

$-$

x_2

$+$

$4x_3$

$+$

$3x_4$

$=$

4

Forward Elimination:

$E'_2 = E_2 - (a_{21}/a_{11}) \times E_1:$

$-\frac{2}{1} \times$

$($

$2x_1$

$-$

$2x_2$

$+$

$3x_3$

$-$

$3x_4$

$=$

-20

$)$

$($

x_1

$-$

x_2

$+$

$2x_3$

$-$

x_4

$=$

-8

$)$

$-$

x_3

$-$

x_4

$=$

-4

$E'_3 = E_3 - (a_{31}/a_{11}) \times E_1:$

$-\frac{1}{1} \times$

$($

x_1

$+$

x_2

$+$

x_3

$=$

-2

$)$

$($

x_1

$-$

x_2

$+$

$2x_3$

$-$

x_4

$=$

-8

$)$

$2x_2$

$-$

x_3

$+$

x_4

$=$

6

$($

x_1

$-$

x_2

$+$

$4x_3$

$+$

$3x_4$

$=$

4

$)$

$E'_4 = E_4 - (a_{41}/a_{11}) \times E_1:$

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$E_1:$

x_1

$-$

x_2

$+$

$2x_3$

$-$

x_4

$=$

-8

$E_2:$

$2x_1$

$-$

$2x_2$

$+$

$3x_3$

$-$

$3x_4$

$=$

-20

$E_3:$

x_1

$+$

x_2

$+$

x_3

$=$

-2

$E_4:$

x_1

$-$

x_2

$+$

$4x_3$

$+$

$3x_4$

$=$

4

Forward Elimination:

$E'_2 = E_2 - (a_{21}/a_{11}) \times E_1:$

$-\frac{2}{1} \times$

$($

$2x_1$

$-$

$2x_2$

$+$

$3x_3$

$-$

$3x_4$

$=$

-20

$)$

$($

x_1

$-$

x_2

$+$

$2x_3$

$-$

x_4

$=$

-8

$)$

$-$

x_3

$-$

x_4

$=$

-4

$E'_3 = E_3 - (a_{31}/a_{11}) \times E_1:$

$-\frac{1}{1} \times$

$($

x_1

$+$

x_2

$+$

x_3

$=$

-2

$)$

$($

x_1

$-$

x_2

$+$

$2x_3$

$-$

x_4

$=$

-8

$)$

$2x_2$

$-$

x_3

$+$

x_4

$=$

6

$E'_4 = E_4 - (a_{41}/a_{11}) \times E_1:$

$-\frac{1}{1} \times$

$($

x_1

$-$

x_2

$+$

$4x_3$

$+$

$3x_4$

$=$

4

$)$

$($

x_1

$-$

x_2

$+$

$2x_3$

$-$

x_4

$=$

-8

$)$

Example 03-01: Find root of the following system of equations.

$$\begin{array}{rcccccccl} E_1: & x_1 & - & x_2 & + & 2x_3 & - & x_4 & = & -8 \\ E_2: & 2x_1 & - & 2x_2 & + & 3x_3 & - & 3x_4 & = & -20 \\ E_3: & x_1 & + & x_2 & + & x_3 & & & = & -2 \\ E_4: & x_1 & - & x_2 & + & 4x_3 & + & 3x_4 & = & 4 \end{array}$$

$$\begin{array}{lcl}
 E'_2 = E_2 - (a_{21}/a_{11}) \times E_1: & -\frac{2}{1} \times & \begin{array}{r}
 \left(\begin{array}{ccccccc} 2x_1 & - & 2x_2 & + & 3x_3 & - & 3x_4 & = & -20 \end{array} \right) \\
 \left(\begin{array}{ccccccc} x_1 & - & x_2 & + & 2x_3 & - & x_4 & = & -8 \end{array} \right) \\
 \hline
 \phantom{\left(\begin{array}{ccccccc} x_1 & - & x_2 & + & 2x_3 & - & x_4 & = & -8 \end{array} \right)} - x_3 - x_4 = -4 \\
 \hline \hline
 \end{array} \\
 E'_3 = E_3 - (a_{31}/a_{11}) \times E_1: & -\frac{1}{1} \times & \begin{array}{r}
 \left(\begin{array}{ccccccc} x_1 & + & x_2 & + & x_3 & & & = & -2 \end{array} \right) \\
 \left(\begin{array}{ccccccc} x_1 & - & x_2 & + & 2x_3 & - & x_4 & = & -8 \end{array} \right) \\
 \hline
 \phantom{\left(\begin{array}{ccccccc} x_1 & - & x_2 & + & 2x_3 & - & x_4 & = & -8 \end{array} \right)} 2x_2 - x_3 + x_4 = 6 \\
 \hline \hline
 \end{array} \\
 E'_4 = E_4 - (a_{41}/a_{11}) \times E_1: & -\frac{1}{1} \times & \begin{array}{r}
 \left(\begin{array}{ccccccc} x_1 & - & x_2 & + & 4x_3 & + & 3x_4 & = & 4 \end{array} \right) \\
 \left(\begin{array}{ccccccc} x_1 & - & x_2 & + & 2x_3 & - & x_4 & = & -8 \end{array} \right) \\
 \hline
 \phantom{\left(\begin{array}{ccccccc} x_1 & - & x_2 & + & 2x_3 & - & x_4 & = & -8 \end{array} \right)} 2x_3 + 4x_4 = 12 \\
 \hline \hline
 \end{array}
 \end{array}$$

Gaussian Elimination Method

Forward Elimination:

$$\begin{array}{lclclclclcl} E_1: & x_1 & - & x_2 & + & 2x_3 & - & x_4 & = & -8 \\ E'_2: & & & & - & x_3 & - & x_4 & = & -4 \\ E'_3: & & & 2x_2 & - & x_3 & + & x_4 & = & 6 \\ E'_4: & & & & & 2x_3 & + & 4x_4 & = & 12 \end{array}$$

Gaussian Elimination Method

Forward Elimination:

$$\begin{array}{lcl} E_1: & x_1 - x_2 + 2x_3 - x_4 = -8 \\ E'_2: & -x_3 - x_4 = -4 \\ E'_3: & 2x_2 - x_3 + x_4 = 6 \\ E'_4: & 2x_3 + 4x_4 = 12 \end{array} \rightarrow \begin{array}{lcl} E_1: & x_1 - x_2 + 2x_3 - x_4 = -8 \\ E'_2: & 2x_2 - x_3 + x_4 = 6 \\ E''_3: & -x_3 - x_4 = -4 \\ E''_4: & 2x_3 + 4x_4 = 12 \end{array}$$

Gaussian Elimination Method

Forward Elimination:

$$\begin{array}{lcl} E_1: & x_1 & - x_2 + 2x_3 - x_4 = -8 \\ E_2': & & - x_3 - x_4 = -4 \\ E_3': & 2x_2 & - x_3 + x_4 = 6 \\ E_4': & & 2x_3 + 4x_4 = 12 \end{array} \quad \rightarrow \quad \begin{array}{lcl} E_1: & x_1 & - x_2 + 2x_3 - x_4 = -8 \\ E_2': & & 2x_2 - x_3 + x_4 = 6 \\ E_3'': & & - x_3 - x_4 = -4 \\ E_4'': & & 2x_3 + 4x_4 = 12 \end{array}$$

$$E_4''' = E_4'' - (a_{43}''/a_{33}'') \times E_3'':$$

Gaussian Elimination Method

Forward Elimination:

$$\begin{array}{rcl}
 E_1: & x_1 & - x_2 + 2x_3 - x_4 = -8 \\
 E_2': & & - x_3 - x_4 = -4 \\
 E_3': & 2x_2 & - x_3 + x_4 = 6 \\
 E_4': & & 2x_3 + 4x_4 = 12
 \end{array}
 \rightarrow
 \begin{array}{rcl}
 E_1: & x_1 & - x_2 + 2x_3 - x_4 = -8 \\
 E_2': & & 2x_2 - x_3 + x_4 = 6 \\
 E_3'': & & - x_3 - x_4 = -4 \\
 E_4'': & & 2x_3 + 4x_4 = 12
 \end{array}$$

($2x_3 + 4x_4 = 12$)

$$E_4''' = E_4'' - (a_{43}''/a_{33}'') \times E_3'':$$

Gaussian Elimination Method

Forward Elimination:

$$\begin{array}{lcl} E_1: & x_1 & - x_2 + 2x_3 - x_4 = -8 \\ E_2': & & - x_3 - x_4 = -4 \\ E_3': & 2x_2 & - x_3 + x_4 = 6 \\ E_4': & & 2x_3 + 4x_4 = 12 \end{array}$$

→

$$\begin{array}{lcl} E_1: & x_1 & - x_2 + 2x_3 - x_4 = -8 \\ E_2': & & 2x_2 - x_3 + x_4 = 6 \\ E_3'': & & - x_3 - x_4 = -4 \\ E_4'': & & 2x_3 + 4x_4 = 12 \end{array}$$

$$-\frac{2}{-1} \times \begin{pmatrix} 2x_3 + 4x_4 = 12 \\ -x_3 - x_4 = -4 \end{pmatrix}$$

$$E_4''' = E_4'' - (a_{43}''/a_{33}'') \times E_3'':$$

Forward Elimination:

$$E_4''' = E_4'' - (a_{43}''/a_{33}'') \times E_3'':$$

Gaussian Elimination Method

Forward Elimination:

$$\begin{array}{lcl} E_1: & x_1 & - x_2 + 2x_3 - x_4 = -8 \\ E_2': & & - x_3 - x_4 = -4 \\ E_3': & 2x_2 & - x_3 + x_4 = 6 \\ E_4': & & 2x_3 + 4x_4 = 12 \end{array}$$

$\rightarrow$

$$\begin{array}{lcl} E_1: & x_1 & - x_2 + 2x_3 - x_4 = -8 \\ E_2': & & \color{red}{2x_2} - \color{red}{x_3} + \color{red}{x_4} = \color{red}{6} \\ E_3'': & & - \color{red}{x_3} - \color{red}{x_4} = \color{red}{-4} \\ E_4'': & & 2x_3 + 4x_4 = 12 \end{array}$$

$$E_4''' = E_4'' - (a_{43}''/a_{33}'') \times E_3'':$$

$$\begin{array}{rcl} & (& 2x_3 + 4x_4 = 12) \\ \color{red}{-\frac{2}{-1}} \times & (& - x_3 - x_4 = -4) \\ \hline & & 2x_4 = 4 \\ \hline \hline \end{array}$$

$$\begin{array}{lcl} E_1: & x_1 & - x_2 + 2x_3 - x_4 = -8 \\ E_2': & & 2x_2 - x_3 + x_4 = 6 \\ E_3'': & & - x_3 - x_4 = -4 \\ E_4''': & & 2x_4 = 4 \end{array}$$

Gaussian Elimination Method

Forward Elimination:

$E_1:$

x_1

$-$

x_2

$+$

$2x_3$

$-$

x_4

$=$

-8

$E'_2:$

$-$

x_3

$-$

x_4

$=$

-4

$E'_3:$

$2x_2$

$-$

x_3

$+$

x_4

$=$

6

$E'_4:$

$2x_3$

$+$

$4x_4$

$=$

12

$\rightarrow$

$E_1:$

x_1

$-$

x_2

$+$

$2x_3$

$-$

x_4

$=$

-8

$E'_2:$

$2x_2$

$-$

x_3

$+$

x_4

$=$

6

$E''_3:$

$-$

x_3

$-$

x_4

$=$

-4

$E''_4:$

$2x_3$

$+$

$4x_4$

$=$

12

$$E'''_4 = E''_4 - (a''_{43}/a''_{33}) \times E''_3:$$

$-\frac{2}{-1} \times$

$($

$2x_3$

$+$

$4x_4$

$=$

12

$)$

$($

$-$

x_3

$-$

x_4

$=$

-4

$)$

$2x_4$

$=$

4

$E_1:$

x_1

$-$

x_2

$+$

$2x_3$

$-$

x_4

$=$

-8

$E'_2:$

$2x_2$

$-$

x_3

$+$

x_4

$=$

6

$E''_3:$

$-$

x_3

$-$

x_4

$=$

-4

$E'''_4:$

$2x_4$

$=$

4

Back Substitution:

Gaussian Elimination Method

Forward Elimination:

$$\begin{array}{lcl} E_1: & x_1 & - x_2 + 2x_3 - x_4 = -8 \\ E_2': & & - x_3 - x_4 = -4 \\ E_3': & 2x_2 & - x_3 + x_4 = 6 \\ E_4': & & 2x_3 + 4x_4 = 12 \end{array}$$

$\rightarrow$

$$\begin{array}{lcl} E_1: & x_1 & - x_2 + 2x_3 - x_4 = -8 \\ E_2': & & \color{red}{2x_2} - \color{red}{x_3} + \color{red}{x_4} = \color{red}{6} \\ E_3'': & & - \color{red}{x_3} - \color{red}{x_4} = \color{red}{-4} \\ E_4'': & & 2x_3 + 4x_4 = 12 \end{array}$$

$$E_4''' = E_4'' - (a_{43}''/a_{33}'') \times E_3'':$$

$$\begin{array}{rcl} & (& 2x_3 + 4x_4 = 12) \\ \color{red}{-\frac{2}{-1}} \times & (& - x_3 - x_4 = -4) \\ \hline & & 2x_4 = 4 \\ \hline \hline \end{array}$$

$$\begin{array}{lcl} E_1: & x_1 & - x_2 + 2x_3 - x_4 = -8 \\ E_2': & 2x_2 & - x_3 + x_4 = 6 \\ E_3'': & & - x_3 - x_4 = -4 \\ E_4''': & & 2x_4 = 4 \end{array}$$

Back Substitution:

$$\color{red}{x_4} = \frac{4}{2} = \color{red}{2}$$

Gaussian Elimination Method

Forward Elimination:

$$\begin{array}{lcl}
 E_1: & x_1 & - x_2 + 2x_3 - x_4 = -8 \\
 E'_2: & & - x_3 - x_4 = -4 \\
 E'_3: & 2x_2 & - x_3 + x_4 = 6 \\
 E'_4: & & 2x_3 + 4x_4 = 12
 \end{array}
 \rightarrow
 \begin{array}{lcl}
 E_1: & x_1 & - x_2 + 2x_3 - x_4 = -8 \\
 E'_2: & & 2x_2 - x_3 + x_4 = 6 \\
 E''_3: & & - x_3 - x_4 = -4 \\
 E''_4: & & 2x_3 + 4x_4 = 12
 \end{array}$$

$$E'''_4 = E''_4 - (a''_{43}/a''_{33}) \times E''_3:$$

$$\begin{array}{rcl}
 & (& 2x_3 + 4x_4 = 12) \\
 -\frac{2}{-1} \times & (& - x_3 - x_4 = -4) \\
 \hline
 & & 2x_4 = 4 \\
 \hline \hline
 \end{array}$$

$$\begin{array}{lcl}
 E_1: & x_1 & - x_2 + 2x_3 - x_4 = -8 \\
 E'_2: & & 2x_2 - x_3 + x_4 = 6 \\
 E''_3: & & - x_3 - x_4 = -4 \\
 E'''_4: & & 2x_4 = 4
 \end{array}$$

Back Substitution:

$$\begin{aligned}
 x_4 &= \frac{4}{2} = 2 \\
 x_3 &= \frac{-4 - (-x_4)}{-1} = \frac{-4 - (-2)}{-1} = 2
 \end{aligned}$$

Gaussian Elimination Method

Forward Elimination:

$$\begin{array}{rcll}
 E_1: & x_1 & - & x_2 + 2x_3 - x_4 = -8 \\
 E'_2: & & & -x_3 - x_4 = -4 \\
 E'_3: & 2x_2 & - & x_3 + x_4 = 6 \\
 E'_4: & & & 2x_3 + 4x_4 = 12
 \end{array}
 \rightarrow
 \begin{array}{rcll}
 E_1: & x_1 & - & x_2 + 2x_3 - x_4 = -8 \\
 E'_2: & & & 2x_2 - x_3 + x_4 = 6 \\
 E''_3: & & & -x_3 - x_4 = -4 \\
 E''_4: & & & 2x_3 + 4x_4 = 12
 \end{array}$$

$$E'''_4 = E''_4 - (a''_{43}/a''_{33}) \times E''_3:$$

$$\begin{array}{r}
 (\quad 2x_3 + 4x_4 = 12 \quad) \\
 -\frac{2}{-1} \times (\quad -x_3 - x_4 = -4 \quad) \\
 \hline
 \quad \quad \quad 2x_4 = 4 \\
 \hline \hline
 \end{array}$$

$$\begin{array}{rcll}
 E_1: & x_1 & - & x_2 + 2x_3 - x_4 = -8 \\
 E'_2: & & & 2x_2 - x_3 + x_4 = 6 \\
 E''_3: & & & -x_3 - x_4 = -4 \\
 E'''_4: & & & 2x_4 = 4
 \end{array}$$

Back Substitution:

$$\begin{aligned}
 x_4 &= \frac{4}{2} = 2 \\
 x_3 &= \frac{-4 - (-x_4)}{-1} = \frac{-4 - (-2)}{-1} = 2 \\
 x_2 &= \frac{6 - (-x_3) - x_4}{2} = \frac{6 - (-2) - 2}{2} = 3
 \end{aligned}$$

Gaussian Elimination Method

Forward Elimination:

$$\begin{array}{lcl} E_1: & x_1 & - x_2 + 2x_3 - x_4 = -8 \\ E'_2: & & - x_3 - x_4 = -4 \\ E'_3: & 2x_2 & - x_3 + x_4 = 6 \\ E'_4: & & 2x_3 + 4x_4 = 12 \end{array}$$

→

$$\begin{array}{lcl} E_1: & x_1 & - x_2 + 2x_3 - x_4 = -8 \\ E'_2: & & 2x_2 - x_3 + x_4 = 6 \\ E''_3: & & - x_3 - x_4 = -4 \\ E''_4: & & 2x_3 + 4x_4 = 12 \end{array}$$

$$E'''_4 = E''_4 - (a''_{43}/a''_{33}) \times E''_3:$$

$$\begin{array}{r} \left(\begin{array}{ccc} 2x_3 & + & 4x_4 = 12 \end{array} \right) \\ -\frac{2}{-1} \times \left(\begin{array}{ccc} - & x_3 & - x_4 = -4 \end{array} \right) \\ \hline \begin{array}{ccc} & 2x_4 & = 4 \end{array} \\ \hline \hline \end{array}$$

$$\begin{array}{lcl} E_1: & x_1 & - x_2 + 2x_3 - x_4 = -8 \\ E'_2: & & 2x_2 - x_3 + x_4 = 6 \\ E''_3: & & - x_3 - x_4 = -4 \\ E'''_4: & & 2x_4 = 4 \end{array}$$

Back Substitution:

$$\begin{aligned} x_4 &= \frac{4}{2} = 2 \\ x_3 &= \frac{-4 - (-x_4)}{-1} = \frac{-4 - (-2)}{-1} = 2 \\ x_2 &= \frac{6 - (-x_3) - x_4}{2} = \frac{6 - (-2) - 2}{2} = 3 \\ x_1 &= \frac{-8 - (-x_2) - 2x_3 - (-x_4)}{1} = \frac{-8 - (-3) - 2 \cdot (2) - (-2)}{1} = -7 \end{aligned}$$